**STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT SYSTEM**

**A PROJECT REPORT**

Submitted by

**THARUN B P**

**(2116241501230)**

in partial fulfilment of the End Semester Examination for the course

**CB23F35 –**

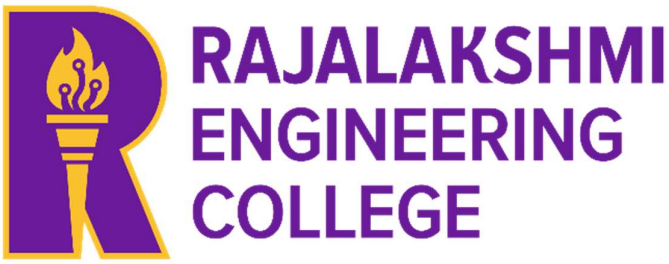
**LOW-CODE APPLICATION DEVELOPMENT**

**USING ZOHO CREATOR**

in

BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning




RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

2026-2027

# BONAFIDE CERTIFICATE

Certified that this project report titled "STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT SYSTEM" is the bonafide work of THARUN B P (2116241501230), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

**SIGNATURE**

Dr Ayyadurai M
HEAD OF THE DEPARTMENT
Department of Artificial Intelligence
And Machine Learning
Rajalakshmi Engineering College
Thandalam – 602 105

**SIGNATURE**

Mrs Arulmozhi C
COURSE FACULTY
Assistant Professor
Department of Artificial Intelligence
And Machine Learning
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on Not evidenced.

**INTERNAL EXAMINER**

**EXTERNAL EXAMINER**

# DECLARATION

I, THARUN B P (241501209), hereby declare that the project report "STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT SYSTEM" submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me. The application was configured using Zoho Creator and its AI-assisted builder. Any AI-assisted features, code snippets, templates, or external resources used are to be acknowledged appropriately.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit.

Place:Chennai
Date: 19 October 2026

SIGNATURE OF THE STUDENT
THARUN B P
241501230

## ACKNOWLEDGEMENT

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I sincerely thank my course faculty, Mrs. Arulmozhi C, for the guidance and feedback provided throughout the development of this application and preparation of this report.

I also acknowledge Zoho Creator and its learning resources for providing a low-code environment in which the application could be configured. Finally, I thank my family and friends for their support.

THARUN B P

# ABSTRACT

Student internship and placement activities involve students, companies, internship opportunities, job openings, applications, interviews, offers, and internship completion records. When these activities are handled through separate spreadsheets or informal communication, information can become fragmented, duplicate entries may occur, application statuses can be difficult to track, and preparing summaries may require repeated manual effort. The Student Internship and Placement Management System was developed as a low-code application using Zoho Creator to provide a central place to record and review these activities.

The application contains forms for student registration, company registration, internship opportunities, job openings, internship applications, placement applications, interview schedules, placement offers, and internship completion. Reports were created to display the main records, and a dashboard was configured with six summary cards and three pie charts. Eight-plus workflows were reported as created and enabled by the Zoho Creator Zia Build Agent, including default-status workflows and validation workflows for internship applications, placement applications, and placement offers. The validation scripts and field mappings must be checked in the live application and tested with actual records before their behaviour is treated as verified. No external integration, production deployment, or role-based access configuration is claimed in this report unless later confirmed with evidence.

Development was carried out through the Zoho Creator application builder and its AI-assisted configuration facilities. The current build has been reported to contain the main forms, reports, dashboard, and workflows. The next validation step is end-to-end manual testing: adding sample records, confirming that related records appear in reports, testing duplicate application prevention and invalid salary rejection, and checking that dashboard values update from saved data. Actual performance measurements and user-acceptance results have not yet been collected; therefore, quantitative improvements are not claimed. The application provides a practical foundation for centralised internship and placement record management and can be extended with role-based access, notifications, deadline reminders, and verified deploymentcontrols.



Keywords: Low-Code Development; Zoho Creator; Student Placement; Internship Management; Workflow Automation; Deluge Scripting.

# PROJECT AT A GLANCE

| Attribute | Details |
|---|---|
| Project Title | Student Internship and Placement Management System |
| Application Domain | Education / Campus Recruitment |
| Student Name & Register No. | THARUN B P – 241501230 |
| Zoho Creator Edition | Student Edition |
| Application Name (Link Name) | student-internship-placement-management-system |
| Published / Live URL | https://creatorapp.zoho.in/tharun.bp.2024.aiml_rajal6/environment/development/student-placement-management-system#Section_1 |
| Demo Credentials | Not included in this report. |
| Number of Forms / Subforms | 9 main forms |
| Number of Reports (by type) | 9 reports |
| Number of Pages / Dashboards | 1 dashboard: Student_Placement_Dashboard |
| Number of Workflows | 9 workflows |
| Deluge Functions / Custom Actions | 3 validation workflows |
| Integrations | No external integrations configured |
| Roles & Profiles Defined | Administrator / placement-management stakeholders |
| Environments Used | Zoho Creator Student Edition |
| Date of Production Deployment | Public Company Registration Form link available; formal deployment date not recorded |

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## LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| API | Application Programming Interface |
| CRUD | Create, Read, Update, Delete |
| ER | Entity–Relationship |
| JSON | JavaScript Object Notation |
| LCAP | Low-Code Application Platform |
| RBAC | Role-Based Access Control |
| REST | Representational State Transfer |
| SDLC | Software Development Life Cycle |
| UAT | User Acceptance Testing |
| UI | User Interface |
| Zia | Zoho's AI assistant |
| Deluge | Data Enriched Language for the Universal Grid Environment |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

Internship and placement coordination is an important activity in an engineering college. It involves maintaining student profiles, company details, internship opportunities, job openings, applications, interview schedules, offers, and internship completion records. These activities are connected: a student applies to an opportunity, the application is reviewed, interviews may be scheduled, an offer may be issued, and an internship may later be marked complete.

When information is maintained in separate spreadsheets, documents, and messages, staff may need to search multiple locations to answer simple questions such as which students applied to a role, which interviews are upcoming, or which offers have been accepted. A centralised application can provide structured data entry, searchable reports, status fields, and a dashboard for quick overview.

This project uses Zoho Creator, a low-code application platform, to build a Student Internship and Placement Management System. The application was configured with forms, reports, a dashboard, and record-creation workflows. The project focuses on a simple working prototype rather than claiming a fully integrated enterprise recruitment platform.

## 1.2 Problem Statement

The problem addressed is the lack of a single structured place to maintain and review internship and placement information from student registration through applications, interviews, offers, and internship completion. Disconnected manual records can create duplicated data, inconsistent statuses, delayed searches, and difficulty preparing current summaries.

- Student, company, opportunity, and application details may be scattered across multiple files or messages.
- Duplicate applications or incomplete records may not be detected consistently without validation.
- Upcoming interviews and application statuses may require manual follow-up.

- Preparing summaries for students, companies, applications, and offers can require repeated manual work.
- Without verified access permissions, private student information may be visible to users who should not access it.

### 1.3 Objectives

- To create structured forms for students, companies, internship opportunities, and job openings.
- To record and track internship and placement applications using status fields.
- To maintain interview schedules, placement offers, and internship completion records.
- To create reports that allow users to view records by module.
- To provide a dashboard with summary cards and charts based on application data.
- To configure simple workflows and validation rules, then test them using actual records.

### 1.4 Scope of the Project

In scope: student and company registration; internship and job opportunity management; internship and placement applications; interview scheduling; placement offers; internship completion; reports; a summary dashboard; default-status workflows; and validation logic reported by the builder.

Out of scope in the current build unless separately configured: external job-board integration, email/SMS notifications, payment processing, predictive candidate ranking, production deployment across separate environments, verified role-based record restrictions, and a student self-service portal. The existence of forms and workflows does not by itself prove these advanced features are available or enabled.

### 1.5 Justification for a Low-Code Approach

Zoho Creator was selected because the application is mainly a record-management workflow. Forms, lookup fields, reports, dashboard components, and workflow rules can be configured through a visual builder, while Deluge can support custom validation logic. This reduces the amount of application infrastructure that must be written manually and makes routine changes easier to maintain. The choice is

appropriate for a student prototype, although platform limits and the need to verify generated scripts remain important.

Zoho Creator's official documentation and Deluge reference are used as technical starting points [1], [2].

### 1.6 Organisation of the Report

Chapter 2 studies the existing process and low-code approach. Chapter 3 presents stakeholders and requirements. Chapter 4 describes architecture, data entities, forms, workflows, navigation, and access control design. Chapter 5 documents the configured application. Chapter 6 defines testing and debugging. Chapter 7 discusses deployment, security, and governance. Chapter 8 reviews outcomes and limitations. Chapter 9 concludes the work and lists future enhancements.

# CHAPTER 2

# LITERATURE SURVEY AND SYSTEM STUDY

## 2.1 Study of the Existing System

A typical manual workflow begins when student information is collected, followed by the recording of companies and available opportunities. Students apply for internships or jobs, and staff review applications and coordinate interviews. Selection outcomes are then communicated and offers are tracked. Internship completion details and certificates are recorded later. In a spreadsheet-based process, these steps can be stored in different files and updated by different people.

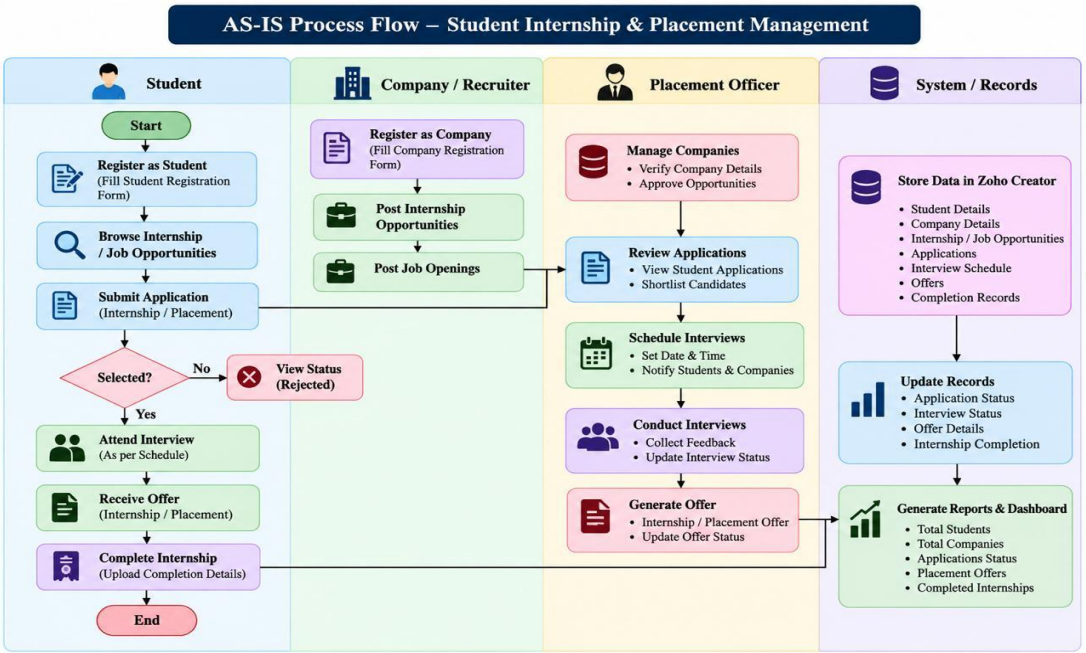



**Figure 2.1: As-Is Process Flow of the Existing System (replace with a simple flow diagram)**

## 2.2 Limitations of the Existing System

- Data is fragmented across separate files and communication channels.
- Searching and preparing reports can require manual consolidation.
- Status changes may not be reflected consistently in all copies of a record.
- Duplicate applications and invalid values may pass without specific checks.
- A manual process makes it harder to view a consolidated dashboard.

### 2.3 Low-Code versus Traditional Development

**Table 2.1: Comparison of Low-Code and Traditional Development**

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| Development effort | Requires coding of data model, UI, server logic, deployment and maintenance. | Forms, reports, menus and workflows can be configured visually; custom logic may use Deluge. |
| Skills required | Programming, database, frontend, backend and deployment skills. | Platform configuration skills and Deluge knowledge for advanced logic. |
| Hosting and infrastructure | Must be selected and managed by the development team. | Platform-hosted runtime is provided by Zoho Creator, subject to the account plan. |
| Reporting | Reports and dashboards are developed or integrated separately. | Built-in report and dashboard builders are available. |
| Maintenance | Changes may require code updates and redeployment. | Many field, report and workflow changes can be made in the builder. |
| Customisation | High flexibility, with higher implementation effort. | Convenient within platform capabilities; specialised UI or logic may require extensions. |
| Cost | Depends on development effort and infrastructure. | Depends on the Creator edition, usage and implementation requirements. |

### 2.4 Literature Survey

The following sources provide platform documentation and background for the implementation. The report does not claim a systematic academic review of five research papers; add verified papers from the college library if the course faculty requires a research-focused survey.

**Table 2.2: Summary of Literature Survey**

| S.No. | Source / Year | Title / Product | Key Focus | Relevance |
|---|---|---|---|---|
| 1 | Zoho Corporation | Zoho Creator Help Documentation | Forms, reports, pages and application configuration. | Primary platform guidance for building the application. |
| 2 | Zoho Corporation | Deluge Script Referenc | Deluge syntax, conditions and data operations. | Reference for validation and |

| | | | | |
|---|---|---|---|---|
| | | e | | workflow logic. |
| 3 | Zoho Corporation | Zoho Creator Resources | Learning material and product capabilities. | Supports understanding of low-code features. |
| 4 | Zoho Corporation | Zoho Creator Workflow documentation | Record events and workflow actions. | Relevant to status-setting automation. |
| 5 | Zoho Corporation | Zoho Creator security / permissions documentation | User permissions and access configuration. | Relevant to protecting student and placement records. |

### 2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly combine visual application design with reusable components, workflow automation, analytics, and AI-assisted configuration. In this project, the Zia Build Agent was used to help configure forms, reports, dashboard components, and workflows. Generated configuration should still be inspected and tested; an AI-generated workflow description is not sufficient evidence that the underlying Deluge logic enforces the intended rule.

### 2.6 Proposed System

The proposed system consolidates core internship and placement records in one Zoho Creator application. Its nine forms support data capture, nine reports provide record views, and the Student Placement Dashboard provides six summary cards and three pie charts. Record-creation workflows set default statuses, while three validation workflows were reported to contain custom logic for internship applications, placement applications, and placement offers. End-to-end tests are still required to confirm relationships, duplicate prevention, salary validation, and chart calculations.

# CHAPTER 3

# REQUIREMENT ANALYSIS

## 3.1 Stakeholders and User Roles

**Table 3.1: Stakeholders and User Roles**

| Role | Description | Key Responsibilities |
|---|---|---|
| Administrator / App Owner | Maintains the Creator application. | Manage forms, reports, workflows, and configuration. |
| Placement Officer | Coordinates campus recruitment records. | Maintain students, companies, opportunities, applications, interviews, and offers. |
| Recruiter / Company Contact | Provides recruitment-related information where access is granted. | Review assigned applications and provide interview feedback, if access is configured. |
| Student | Person whose details and applications are stored. | Student self-service access is a future enhancement unless a portal and permissions are configured. |

The roles above describe the intended stakeholders. Actual Zoho Creator user roles and record-level restrictions have not been confirmed and must not be assumed from the menu sections alone.

## 3.2 Functional Requirements

**Table 3.2: Functional Requirements**

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-01 | The system shall capture student registration records. | Placement Officer / Administrator | High |
| FR-02 | The system shall capture company details. | Placement Officer / Administrator | High |
| FR-03 | The system shall store internship opportunities and job openings. | Placement Officer / Administrator | High |

| | | | |
|---|---|---|---|
| FR-04 | The system shall record internship applications and their statuses. | Placement Officer / Administrator | High |
| FR-05 | The system shall record placement applications and their statuses. | Placement Officer / Administrator | High |
| FR-06 | The system shall store interview dates, times, modes, and statuses. | Placement Officer / Administrator | High |
| FR-07 | The system shall record placement offers and offer statuses. | Placement Officer / Administrator | High |
| FR-08 | The system shall record internship completion and certificate details. | Placement Officer / Administrator | Medium |
| FR-09 | The system shall provide reports for the main record types. | Authorised users | High |
| FR-10 | The dashboard shall display summary cards and charts from stored data. | Authorised users | Medium |
| FR-11 | Workflows shall set | Application | Medium |

| | | | |
|---|---|---|---|
| | default statuses on record creation. | administrator | |
| FR-12 | Validation logic shall reject configured invalid application or offer records. | Application administrator / users | High |

### 3.3 Non-Functional Requirements

**Table 3.3: Non-Functional Requirements**

| NFR ID | Category | Requirement |
|---|---|---|
| NFR-01 | Usability | Forms and reports should be understandable and easy to navigate. |
| NFR-02 | Data integrity | Required fields, dropdown values and validation logic should reduce incomplete or invalid records. |
| NFR-03 | Performance | Reports and dashboard components should load within an acceptable time; measure this during testing. |
| NFR-04 | Security | Only authorised users should access private student records; permissions must be configured and tested. |
| NFR-05 | Maintainability | Form, report and workflow names should be consistent and changes documented. |
| NFR-06 | Availability | The application should be accessible when the Zoho Creator service and user connection are available. |
| NFR-07 | Auditability | Record changes and workflow outcomes should be reviewed using available platform logs where configured. |

### 3.4 Hardware and Software Requirements

**Table 3.4: Hardware and Software Requirements**

| Category | Specification |
|---|---|
| Platform | Zoho Creator (edition to be confirmed from account settings) |
| Scripting | Deluge for custom workflow and validation logic, where used |
| Browser | Current version of a supported web browser |
| Client hardware | Laptop or desktop with internet connectivity |
| Additional tools | Zoho Creator Zia Build Agent was used for assisted configuration |
| External services | No API or external integration confirmed in the current build |

### 3.5 Use Case Model

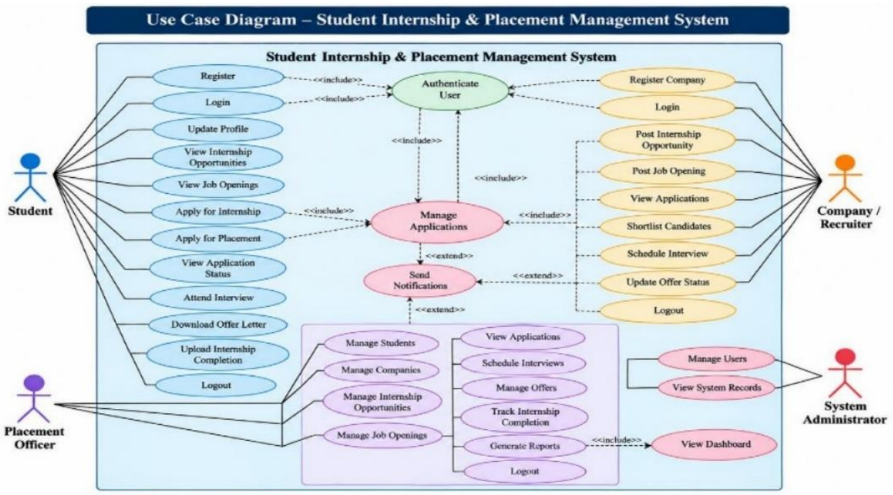



**Figure 3.1: Use Case Diagram of the Application**

**Table 3.5: Use Case Description – Record an Internship Application**

| Item | Description |
|---|---|
| Use Case ID | UC-01 |
| Use Case Name | Record an Internship Application |
| Actor(s) | Placement Officer / authorised application user |
| Pre-condition | Student and internship opportunity records are available. |
| Main Flow | **1. Open Internship_Application_Form. 2. Select the student and internship opportunity. 3. Enter application date and upload resume if required. 4. Submit the form. 5. Check the Internship_Applications report.** |
| Alternate Flow | If validation is configured and detects a duplicate or closed opportunity, submission should be rejected with a message. This behaviour must be tested. |
| Post-condition | A valid application record should be saved and visible in the corresponding report. |
| Related FR | FR-04, FR-09, FR-12 |

# CHAPTER 4

# SYSTEM DESIGN

## 4.1 System Architecture

The application follows a platform-hosted low-code architecture. An authorised user accesses the Zoho Creator application through a web browser. The application layer contains forms, reports, dashboard components, and workflows. Data entered through forms is stored in the Creator application's data layer. Workflows run on configured record events. No separate external API layer is included in the current design because no integration has been confirmed.

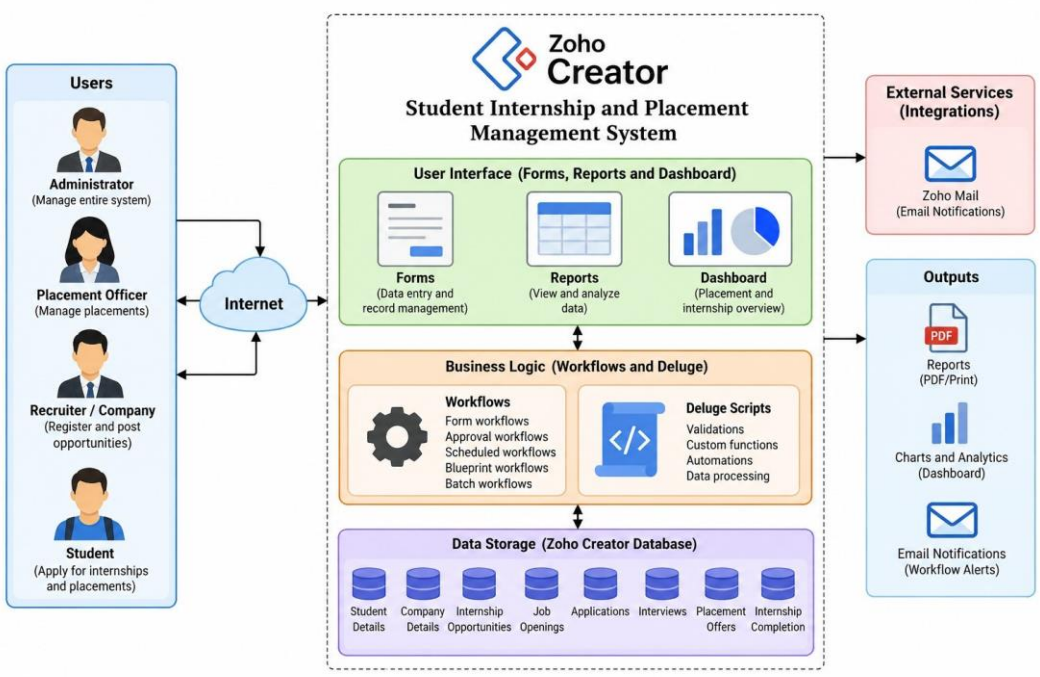



**Figure 4.1: System Architecture of the Application**

## 4.2 Data Model

The main entities are Student, Company, Internship Opportunity, Job Opening, Internship Application, Placement Application, Interview Schedule, Placement Offer, and Internship Completion. The intended relationships connect applications to students and opportunities, interviews to the relevant candidate and recruitment activity, offers to a student and job application, and completion records to an internship application. The actual implementation of each lookup must be checked in the live form builder.

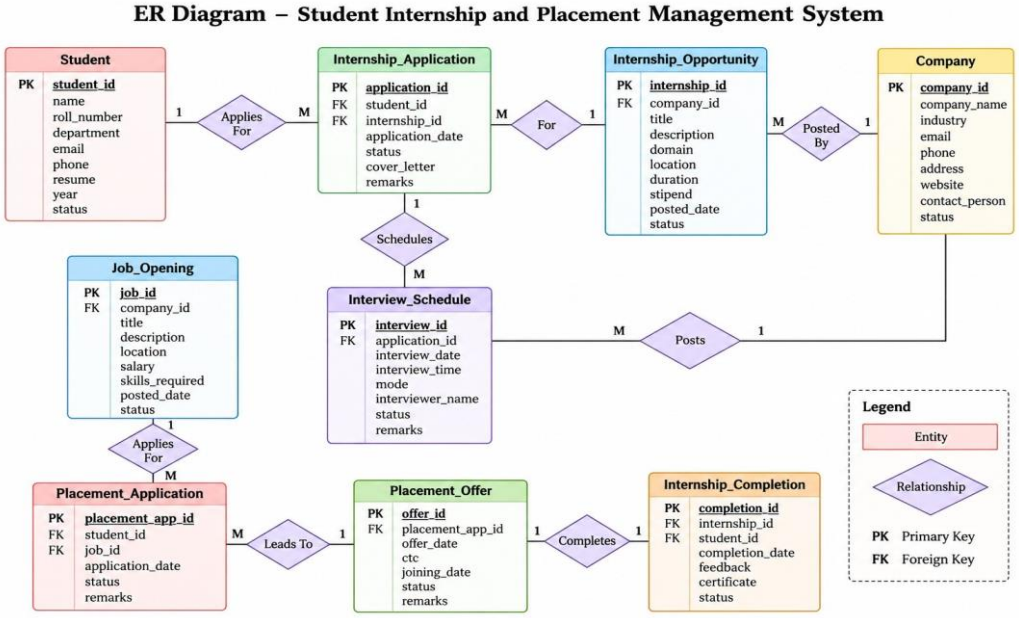



**Figure 4.2: Entity–Relationship Diagram**

**Table 4.1: Entities and Relationships**

| Entity (Form) | Key Information | Related To | Relationship / Implementation |
|---|---|---|---|
| Student_Registration_Form | Student identity, academic and contact details | Applications, interviews, offers, completion | Lookup relationships where configured |
| Company_Registration_Form | Company identity and contact details | Internships, jobs, offers | Lookup relationships where configured |
| Internship_Opportunity_Form | Internship title, eligibility, dates and company | Internship applications | Company lookup reported in initial form |
| Job_Opening_Form | Job title, eligibility, deadline and company | Placement applications and offers | Lookup relationships where configured |
| Internship_Application_Form | Student, opportunity, date, resume, status | Student and internship opportunity | Lookup fields intended; verify live configuration |
| Placement_Application_Form | Student, job, date, resume, status | Student and job opening | Lookup fields intended; verify live configuration |
| Interview_Schedule_Form | Student, company, role, date/time, status, feedback | Application / candidate | Related application linkage may need manual configuration |
| Placement_Offer_Form | Student, company, position, salary, dates, status | Student and placement application | Related lookups intended; verify live configuration |
| Internship_Completion_Form | Student, application, dates, certificate, status | Student and internship application | Related lookups intended; verify live configuration |

4.3

## 4.3 Form Design

The application contains nine main forms. Auto Number fields provide identifiers where configured, dropdowns standardise statuses, file-upload fields support resumes or completion certificates, and multi-line fields capture remarks or feedback. The table below summarises the main form purposes; exact field link names, mandatory flags, and lookup targets should be taken from the live Creator configuration before final submission.

**Table 4.2: Form Field Specification Summary**

| Form Name | Important Fields / Types | Validation or Default Status |
|---|---|---|
| Student_Registration_Form | Student identity and academic details; fields depend on live form | Student status workflow reportedly sets Active |
| Company_Registration_Form | Company name and contact details | Required fields and unique constraints to verify |
| Internship_Opportunity_Form | Company, internship details, dates and eligibility | Company lookup reported; deadline/status fields to verify |
| Job_Opening_Form | Company, job title, dates and eligibility | Open/closed status and deadline fields to verify |
| Internship_Application_Form | Auto Number, student, opportunity, date, resume, status, remarks | Status default Applied; duplicate/deadline validation reported |
| Placement_Application_Form | Auto Number, student, job, date, resume, status, remarks | Status default Applied; duplicate/deadline validation reported |
| Interview_Schedule_Form | Auto Number, student/company or application reference, date/time, mode, status, feedback | Status default Scheduled |
| Placement_Offer_Form | Auto Number, student, company, job application, position, salary, offer/joining dates, status | Status default Pending; salary validation reported |
| Internship_Completion_Form | Auto Number, student, application, start/end dates, certificate, feedback, status | Status default In Progress |

The template asks for a field-by-field specification including field label, link name, field type, mandatory status, validation/default, and remarks. Since the live field link names and mandatory settings were not supplied in the conversation, they are not invented here. Before submission, add one detailed table per form using screenshots or exported field definitions from Zoho Creator.

### 4.4 Process and Workflow Design

The intended process is: register student and company; create internship/job opportunities; record applications; validate required information; schedule interviews; record placement offers; and record internship completion. Workflow triggers reported by Zia are primarily on record creation. Application and offer validation workflows were reported to include custom Deluge logic, but their source code and behaviour must be tested.

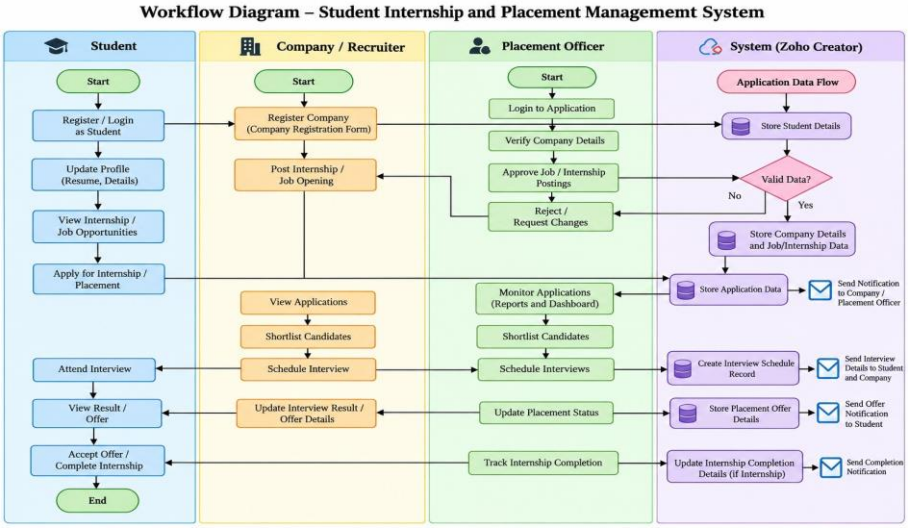



**Figure 4.3: Workflow / Process Flow for Internship and Placement Management**

### 4.5 User Interface and Navigation Design

The application menu is organised into Student Management, Company Management, Opportunity Management, Application Management, Interview Management, Placement Management, and Internship Completion. Reports are available from the menu, and the Student_Placement_Dashboard contains six summary cards and three pie charts.




**Figure 4.4: Navigation Structure and Dashboard Wireframe**

### 4.6 Access Control Design

A permission matrix is included as a design proposal. It does not claim that these roles have been configured in Zoho Creator. The live permissions and record-level restrictions must be reviewed before the system is used with real student information.

**Table 4.3: Role–Permission Matrix (Proposed; verify implementation)**

| Component | Administrator | Placement Officer | Recruiter | Student |
|---|---|---|---|---|
| Student records | Add/Edit/View/ Delete | Add/Edit/View | Assigned records only if configured | Own record only if portal configured |
| Company and opportunity records | Full access | Add/Edit/View | View assigned opportunities if configured | View open opportunities if portal configured |
| Applications | Full access | Manage all | Assigned applications if configured | Own applications if portal configured |
| Interview records | Full access | Manage all | Assigned interviews if configured | Own interviews if portal configured |
| Placement | Full access | Manage all | View assigned | Own records if |

| | | | | |
|---|---|---|---|---|
| offers / completion | | | records if configured | portal configured |
| Dashboard and reports | Full access | View relevant reports | Restricted reports if configured | Own status only if configured |

# CHAPTER 5
# IMPLEMENTATION

## 5.1 Application Setup

The application was built in Zoho Creator with assistance from the Zia Build Agent. The builder was instructed to create the forms first, then reports, dashboard components, and workflows in separate stages to keep configuration manageable. The reported menu sections are Student Management, Company Management, Opportunity Management, Application Management, Interview Management, Placement Management, and Internship Completion.

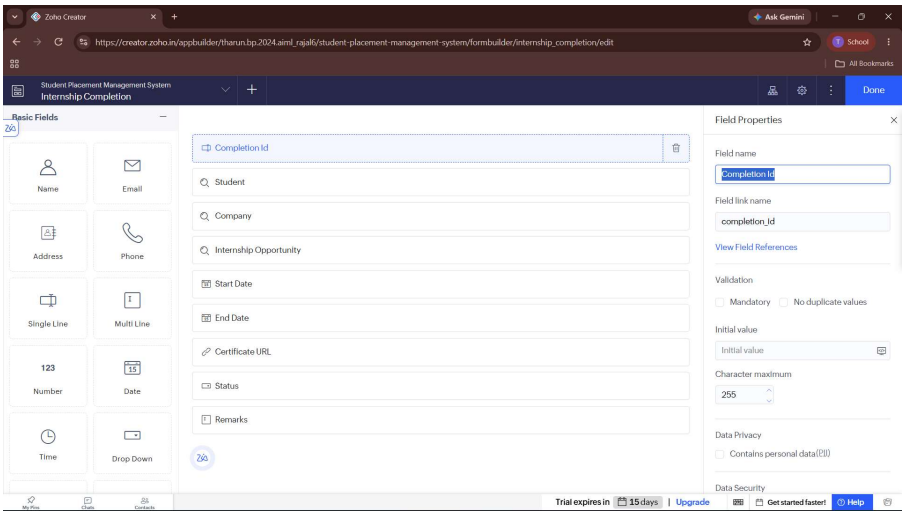



**Figure 5.1: Application Home and Navigation Sections**

## 5.2 Forms and Data Capture

Nine forms are reported as created:Student_Registration_Form, Company_Registration_Form,Internship_Opportunity_Form,Job_Opening_Form, Internship_Application_Form,Placement_Application_Form,Interview_Schedule_Form ,Placement_Offer_Form, and Internship_Completion_Form. They capture the main records required for the recruitment lifecycle. The forms include fields such as identifiers, names, dates, dropdown statuses, resumes, certificates, remarks, and interview feedback, as applicable to each form.

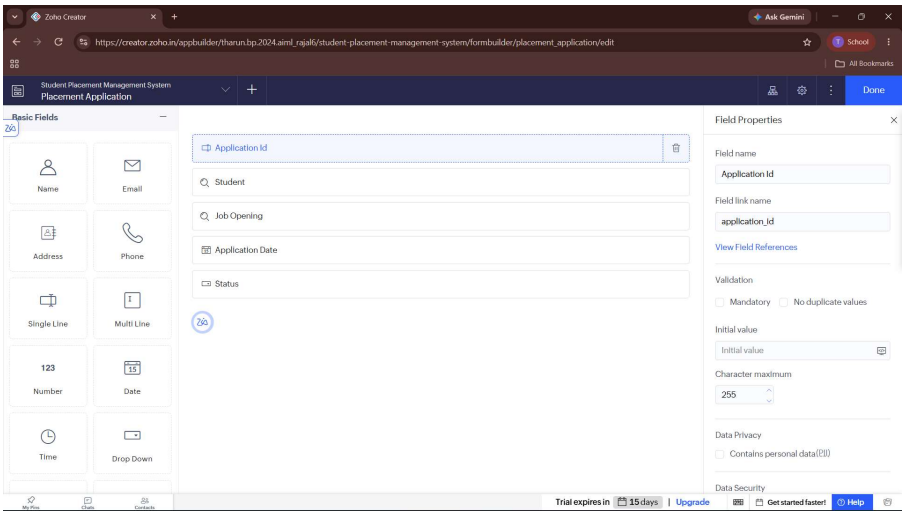



**Figure 5.2: Form Builder Screenshot**

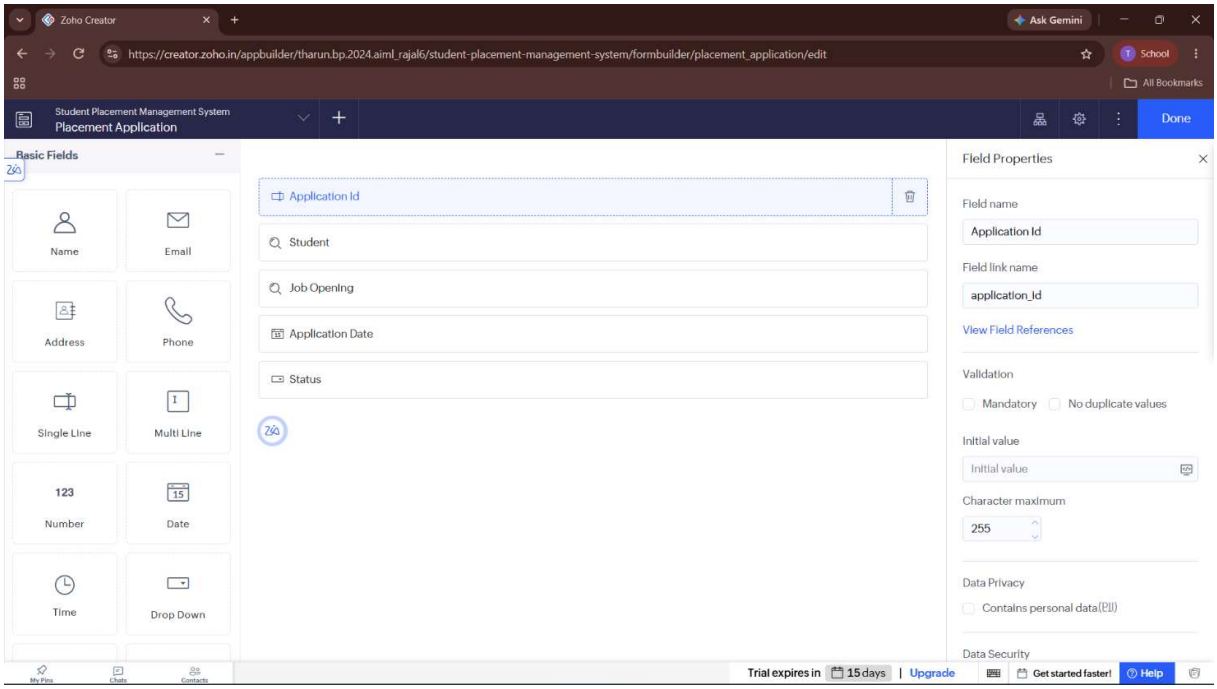



**Figure 5.3: Live Form Screenshot**

### 5.3 Reports

Nine reports were reported as created and added to the menu. The exact report names and report types should be checked in the live app; the names below reflect the confirmed report set from the build conversation.

**Table 5.1: Reports Implemented**

| Report Name | Source Form | Purpose | Status |
|---|---|---|---|
| Student_List | Student_Registration_ Form | View registered students | Reported created; verify live |
| Company_List | Company_Registrati on_For m | View registered companies | Reported created; verify live |
| Internship_List | Internship_Opportun ity_For m | View internship opportuniti es | Reported created; verify live |
| Job_List | Job_Opening_Form | View job openings | Reported created; verify live |
| Internship_Applica tions | Internship_Applicati on_For m | Review internship applications and statuses | Reported created; verify live |
| Placement_Applic ations | Placement_Applicati on_For m | Review job applications and statuses | Reported created; verify live |
| Interview_List | Interview_Schedule_F orm | View interview schedules | Reported created; verify live |
| Placement_Offers | Placement_Offer_For m | Review offers, packages and statuses | Reported created; verify live |
| Internship_Comple tions | Internship_Completi on_For m | View completion records | Reported created; verify live |

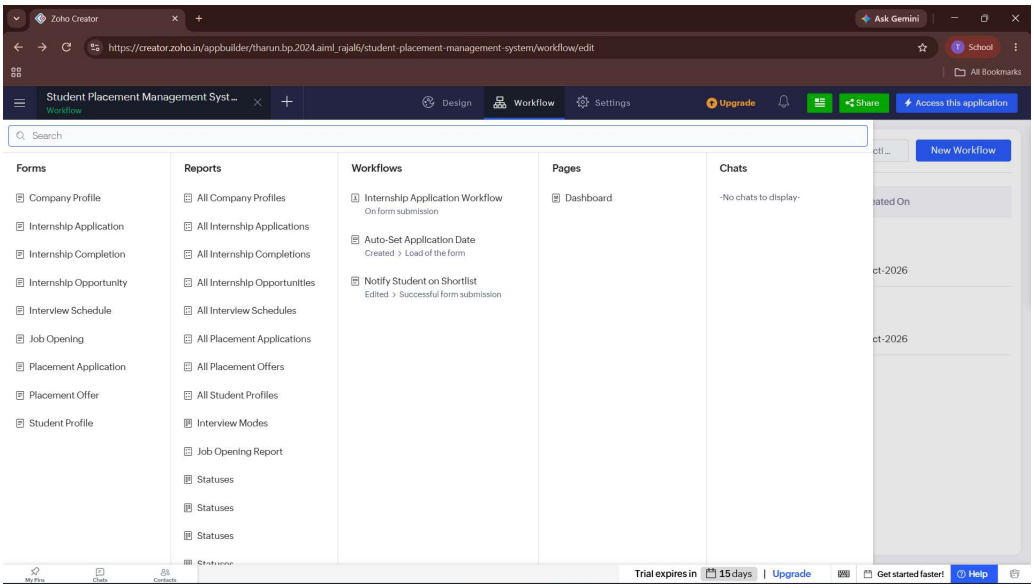



**Figure 5.4: Report Screenshot**

## 5.4 Pages and Dashboards

The Student_Placement_Dashboard was reported as created with six summary cards and three pie charts. The six cards are intended to show total students, total companies, internship applications, placement applications, placement offers, and completed internships. The three charts are intended to show students by department, internship applications by status, and placement applications by status. The values must be verified against actual saved records before claiming that they are correct.

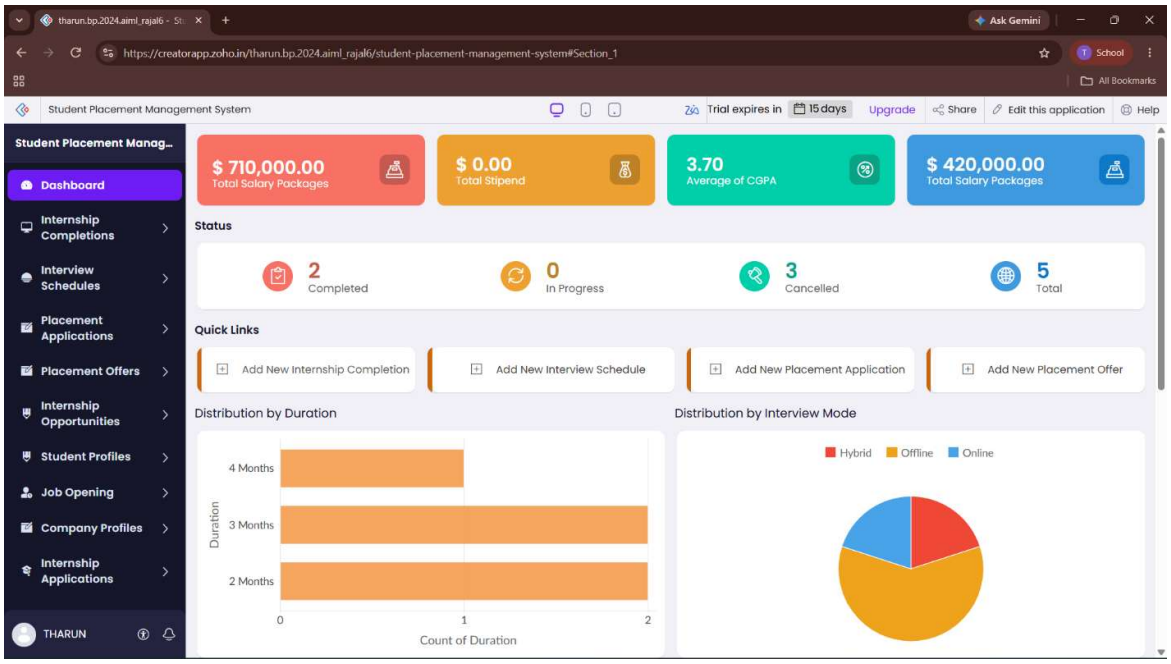



**Figure 5.5 Student Placement Dashboard**

### 5.5 Deluge Scripting

The Zia Build Agent reported adding Deluge logic to three validation workflows: Internship Application Validation, Placement Application Validation, and Placement Offer Validation. The actual source code was not provided in the conversation, so this report does not reproduce guessed scripts. Copy the complete scripts from the Zoho Creator workflow editor into Appendix C and verify their syntax and behaviour.

**Table 5.2: Deluge Scripts Implemented**

| Script ID | Location / Trigger | Purpose | Deluge Concepts |
|---|---|---|---|
| DS-01 | Internship Application Validation / form submission | Check required relationship fields, duplicates, and deadline where implemented | Conditions, record lookup, alert/cancel submit (verify exact code) |
| DS-02 | Placement Application Validation / form submission | Check required relationship fields, duplicates, and deadline where implemented | Conditions, record lookup, alert/cancel submit (verify exact code) |
| DS-03 | Placement Offer Validation / form submission | Check required offer fields and salary greater than zero | Conditions, validation message and submission cancellation (verify exact code) |

**Code 5.1 [validation script – placement offer]**

```
if(input.Student.isNull()){
        alert "Student selection is required";
        cancel submit;
}
if(input.Company.isNull()){
        alert "Company selection is required";
        cancel submit;
}
if(input.Offered_Position.isBlank()){
        alert "Offered Position is required";
        cancel submit;
}
if(input.Joining_Date.isNull()){
        alert "Joining Date is required";
        cancel submit;
}
if(input.Annual_Salary_Package.isNull() ||
input.Annual_Salary_Package <= 0){
        alert "Enter a valid positive salary package";
        cancel submit;
}
```

**Code 5.2 Internship application reminder**

```
studentRecord = Student_Registration_Form[ID == input.Student];
internshipRecord = Internship_Opportunity_Form[ID ==
input.Internship_Opportunity];
studentName = studentRecord.Full_Name.first_name + " " +
studentRecord.Full_Name.last_name;
studentEmail = studentRecord.Email_Address;
internshipTitle = internshipRecord.Internship_Title;
sendmail[
        from : zoho.adminuserid
        to : studentEmail
        subject : "Internship Application Confirmation - " + internshipTitle
        message : "Dear " + studentName + ",<br><br>Your application for
<b>" + internshipTitle + "</b> was submitted
successfully.<br><br>Application Date: " + input.Application_Date +
"<br>Status: " + input.Application_Status + "<br><br>Best
regards,<br>Placement Team"
        content type : HTML]
```

**Script 1 –** Placement Offer Validation: The script validates the placement offer form by checking whether the student, company, offered position, joining date, and salary are provided. It also ensures that the annual salary package is greater than zero. If any field is missing or the salary is invalid, an alert is displayed and submission is cancelled. The result is that only valid placement offers can be submitted.

**Script 2** – Internship Email Notification: The script retrieves the student's name and email address, along with the internship title, application date, and status. It then creates an email containing the application confirmation details and sends it to the student. The result is that the student receives an email confirming successful application submission, provided the workflow and email configuration work correctly.

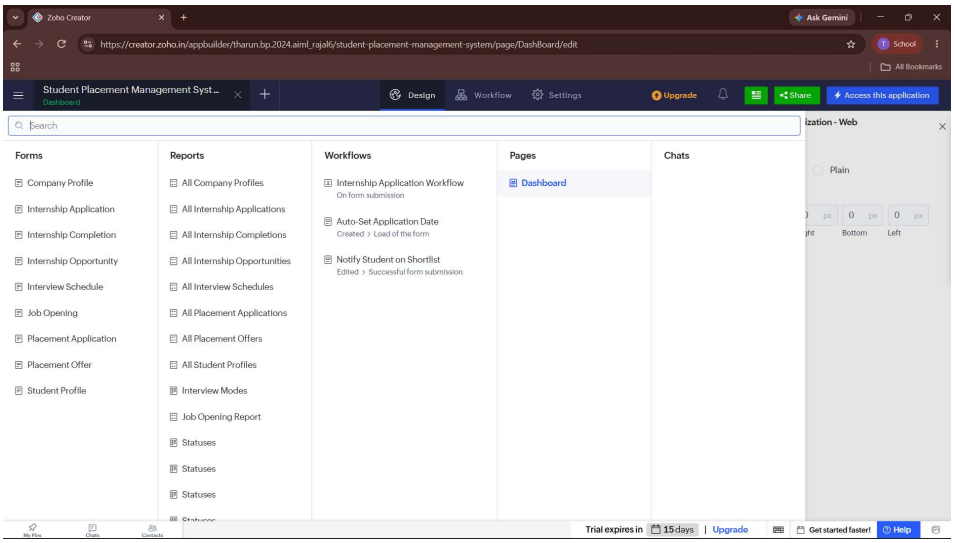
### 5.6 Workflows and Automation

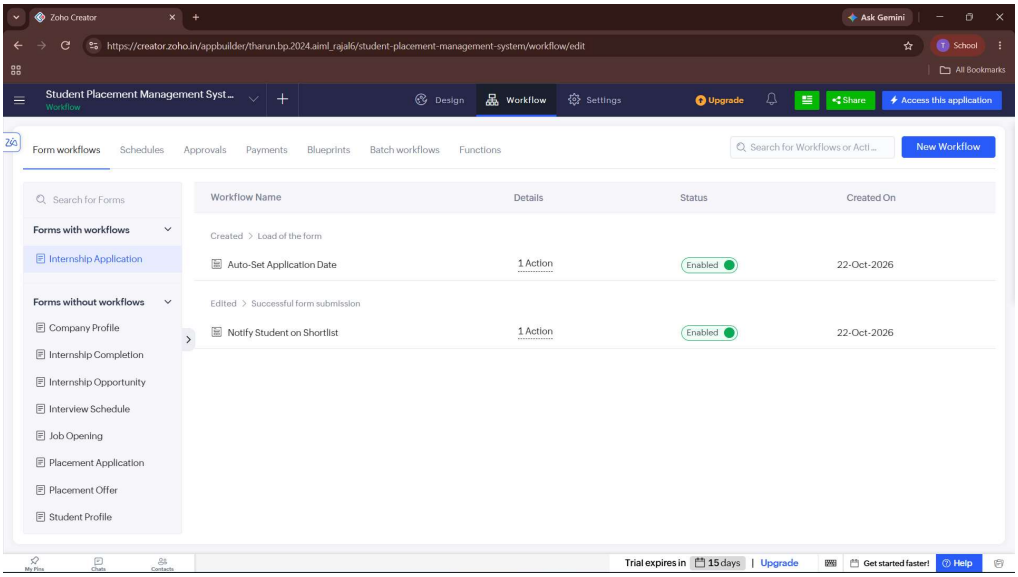
The build conversation reports that the following default-status workflows were created and enabled. Three additional validation workflows were reported as enabled, and custom Deluge logic was later reported as added to those validation workflows. The workflow list in Zoho Creator should be checked for the exact total and trigger/action details.

**Table 5.3: Workflows Implemented**

| Workflow Name | Type | Trigger / Event | Action | Evidence Status |
|---|---|---|---|---|
| Set Student Status to Active | Form workflow | Student record creation | Set status to Active | Reported enabled; test with record |
| Set Internship Application Status to Applied | Form workflow | Internship application creation | Set status to Applied | Reported enabled; test with record |
| Set Placement Application Status to | Form workflow | Placement application creation | Set status to Applied | Reported enabled; test with record |

| | | | | |
|---|---|---|---|---|
| Applied | | | | |
| Set Interview Status to Scheduled | Form workflow | Interview creation | Set status to Scheduled; do not overwrite on edit as reported | Reported enabled; test with record |
| Set Placement Offer Status to Pending | Form workflow | Placement offer creation | Set status to Pending | Reported enabled; test with record |
| Set Internship Completion Status to In Progress | Form workflow | Completion record creation | Set status to In Progress; do not overwrite on edit as reported | Reported enabled; test with record |
| Internship Application Validation | Form workflow | Form submission | Validation logic reportedly added | Inspect code and test negative cases |
| Placement Application Validation | Form workflow | Form submission | Validation logic reportedly added | Inspect code and test negative cases |
| Placement Offer Validation | Form workflow | Form submission | Validation logic reportedly added | Inspect code and test negative cases |


Zoho Creator
https://creator.zoho.in/appbuilder/tharun.bp.2024.aiml_rajal6/student-placement-management-system/page/DashBoard/edit
Student Placement Management Syst...
Dashboard
Design
Workflow
Settings
Upgrade
Share
Access this application
Search
Forms
Company Profile
Internship Application
Internship Completion
Internship Opportunity
Interview Schedule
Job Opening
Placement Application
Placement Offer
Student Profile
Reports
All Company Profiles
All Internship Applications
All Internship Completions
All Internship Opportunities
All Interview Schedules
All Placement Applications
All Placement Offers
All Student Profiles
Interview Modes
Job Opening Report
Statuses
Statuses
Statuses
Workflows
Internship Application Workflow
On form submission
Auto-Set Application Date
Created > Load of the form
Notify Student on Shortlist
Edited > Successful form submission
Pages
Dashboard
Chats
Plain
Bottom
Left
Trial expires in 15 days
Upgrade
Get started faster!
Help


Zoho Creator
https://creator.zoho.in/appbuilder/tharun.bp.2024.aiml_rajal6/student-placement-management-system/workflow/edit
Student Placement Management Syst...
Workflow
Design
Workflow
Settings
Upgrade
Share
Access this application
Form workflows
Schedules
Approvals
Payments
Blueprints
Batch workflows
Functions
Search for Workflows or Acti...
New Workflow
Search for Forms
Forms with workflows
Internship Application
Forms without workflows
Company Profile
Internship Completion
Internship Opportunity
Interview Schedule
Job Opening
Placement Application
Placement Offer
Student Profile
Workflow Name
Details
Status
Created On
Created > Load of the form
Auto-Set Application Date
1 Action
Enabled
22-Oct-2026
Edited > Successful form submission
Notify Student on Shortlist
1 Action
Enabled
22-Oct-2026
Trial expires in 15 days
Upgrade
Get started faster!
Help

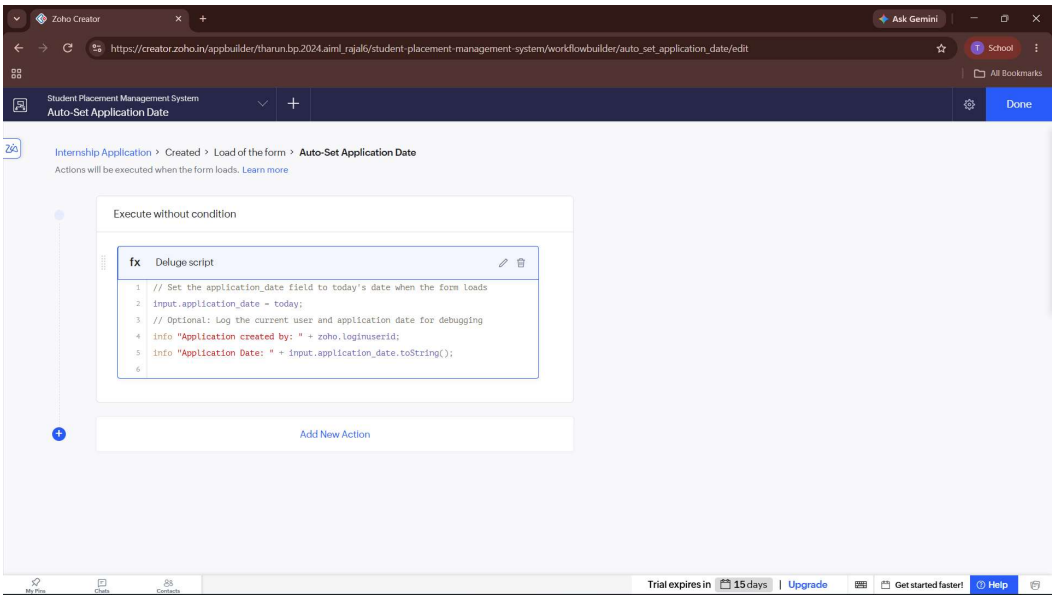



**Figure 5.6: Workflow Configuration**

## 5.7 Integrations

No connection to Zoho CRM, Zoho Mail, a third-party job board, or an external REST API was confirmed in the project build conversation. Therefore, the integration section is retained as required by the template, but no integration is claimed as implemented. An integration could be added in future using a configured connection and secure authentication, with testing evidence.

**Table 5.4: Integrations Implemented**

| 6 Integration | 7 Service / Endpoint | 8 Method | 9 Authentication | 10 Status |
|---|---|---|---|---|
| 11 None confirmed | 12 Not applicable | 13 Not applicable | 14 Not applicable | 15 Not implemented / not verified |

**Code 5.3:[Integration script -Internship Application Confirmation]**

```
studentRecord = Student_Registration_Form[ID == input.Student];
internshipRecord = Internship_Opportunity_Form[ID == input.Internship_Opportunity];
studentEmail = studentRecord.Email_Address;
internshipTitle = internshipRecord.Internship_Title;
sendmail[
   from : zoho.adminuserid
   to : studentEmail
   subject : "Internship Application Confirmation"
```

message : "Your application for " + internshipTitle + " has been submitted successfully."];

Not evidenced

### 5.8 User Management and Role-Based Access

The intended stakeholders include an administrator, placement officer, recruiter, and student. However, the build conversation does not confirm that separate user roles, permission profiles, student portal access, or record-level restrictions were configured. This section is retained because the template requires it. Before claiming RBAC, configure the roles and test access using separate accounts.

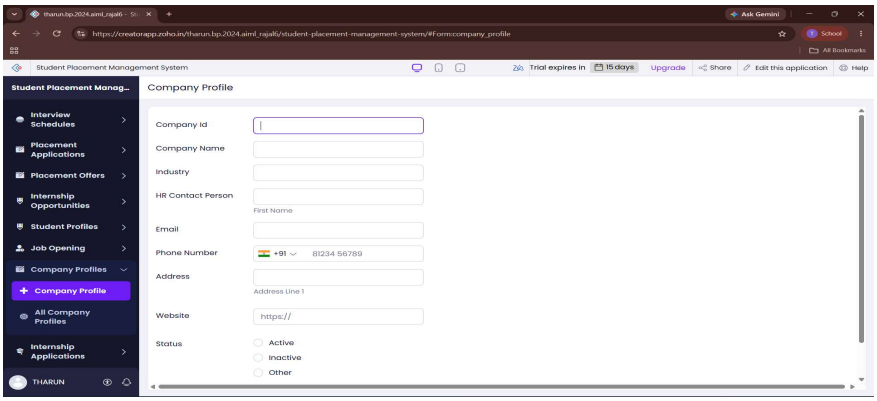
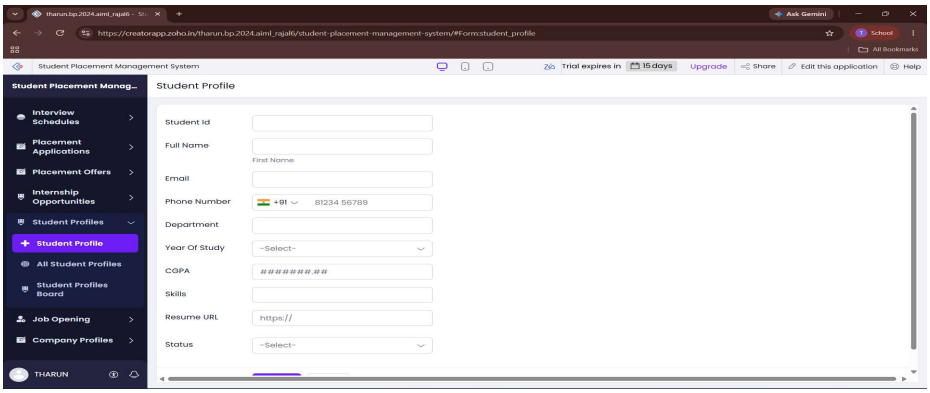



Figure 5.8:Student Role Form and Company Report



## 5.9 AI and Analytics Features

Zia Build Agent was used to assist with creating forms, reports, dashboard components, and workflows. The dashboard contains six summary cards and three pie charts as reported by Zia. AI assistance was used for application configuration; the business rules should be reviewed and tested by the developer. No predictive model, OCR, sentiment analysis, or external Zoho Analytics integration is confirmed.

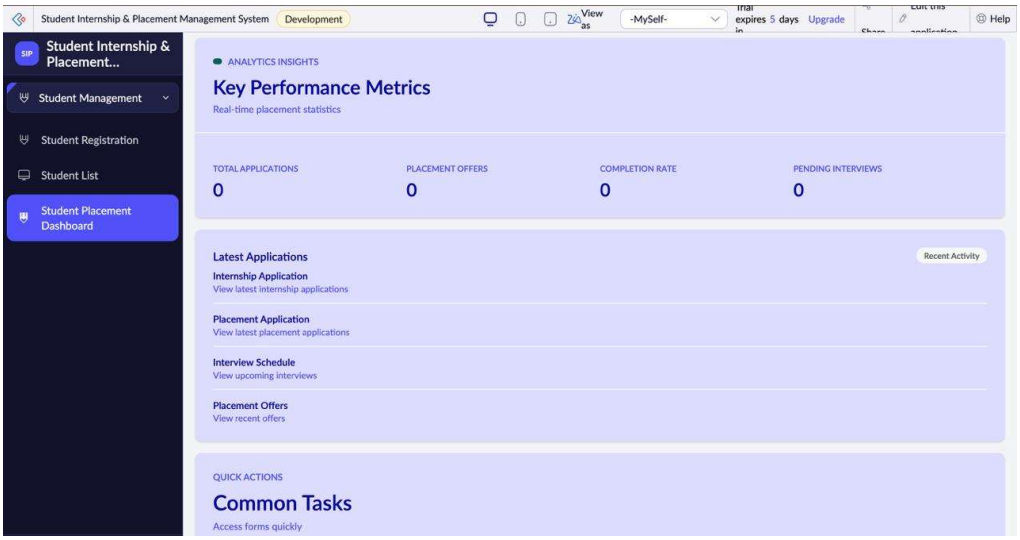



Figure 5.9: Zia / Analytics Feature in Use

## 5.10 Pro-Code Extension

A pro-code extension using Zoho Catalyst, custom JavaScript widgets, or a separate backend service was not used in the current project. The application requirements were addressed through Creator forms, reports, dashboard components, and workflows. A pro-code extension may be considered if future requirements demand complex matching algorithms, advanced analytics, external job-board synchronisation, or custom user experiences that exceed the platform's low-code capabilities.

### 5.11 Mobile and Portal Access

Mobile access and a student-facing portal were not confirmed in the current build. This section is retained to follow the template. If the application is accessed through the Zoho Creator mobile application or a configured portal, insert screenshots and describe the exact access settings. Otherwise, record the feature as not implemented.

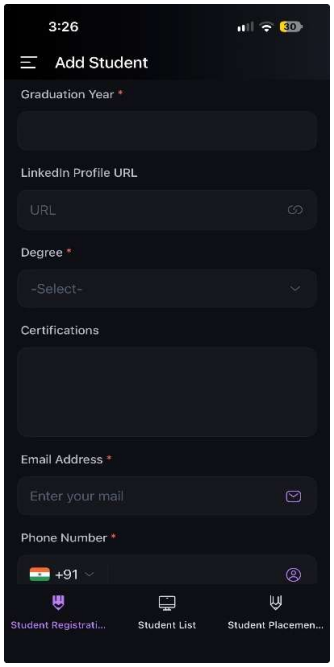



**Figure 5.10: Mobile / Portal Access**

# CHAPTER 6
# TESTING AND DEBUGGING

## 6.1 Testing Strategy

Testing should cover each form, each report, workflow triggers, validation failures, and dashboard calculations. Unit testing checks whether individual forms accept valid records and reject invalid ones. Integration testing checks whether saved records appear in the correct reports and whether dashboard counts change. Negative testing checks duplicate applications and invalid salary values. Role-based testing should be performed only after separate roles and permissions are configured.

At the time this report was prepared, Zia had reported that forms, reports, dashboard, and workflows were created. The conversation did not contain test execution evidence from real sample records. Therefore, the cases below are test plans; update Actual Result and Status after executing them.

## 6.2 Test Cases

**Table 6.1: Test Cases**

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-01 | FR-01 | Create student with required fields | Valid student details | Student record saved | Pending | Pending |
| TC-02 | FR-01 | Omit a mandatory student field | Leave required field blank | Validation message; record not saved | Pending | Pending |
| TC-03 | FR-02 | Create company record | Valid company details | Company record saved | Pending | Pending |
| TC-04 | FR-03 | Create internship opportunity | Valid opportuni ty and | Opportunity saved and listed | Pending | Pending |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | company | | | |
| TC-05 | FR-03 | Create job opening | Valid job and company | Job saved and listed | Pending | Pending |
| TC-06 | FR-04 | Submit valid internship application | Existing student and internship | Application saved and status Applied | Pending | Pending |
| TC-07 | FR-12 | Submit duplicate internship application | Same student and opportuni ty twice | Second submission rejected if duplicate rule is implemente d | Pending | Pending |
| TC-08 | FR-12 | Apply after internship deadline | Deadline in past, if field exists | Submission rejected if deadline rule is implemente d | Pending | Pending |
| TC-09 | FR-05 | Submit valid placement application | Existing student and job | Application saved and status Applied | Pending | Pending |
| TC-10 | FR-12 | Submit duplicate placement application | Same student and job twice | Second submission rejected if duplicate rule is implemente d | Pending | Pending |
| TC-11 | FR-06 | Create | Valid | Interview | Pending | Pending |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | interview record | interview date/time | saved with Scheduled status | | |
| TC-12 | FR-07 | Create placement offer with valid salary | Salary greater than zero | Offer saved with Pending status | Pending | Pending |
| TC-13 | FR-12 | Create placement offer with zero salary | Salary = 0 | Submission rejected if validation is implemented | Pending | Pending |
| TC-14 | FR-08 | Create internship completion record | Valid student/application and dates | Completion saved with In Progress status | Pending | Pending |
| TC-15 | FR-09 | Open all reports | Saved sample records exist | Each report displays its matching records | Pending | Pending |
| TC-16 | FR-10 | Check dashboard summary cards | Known sample records exist | Counts match records in forms/reports | Pending | Pending |
| TC-17 | FR-10 | Check dashboard pie charts | Records have different statuses/departments | Slices reflect actual record counts | Pending | Pending |

| | | | | | | |
|---|---|---|---|---|---|---|
| TC-18 | NFR-04 | Test unauthorised access | Use a non-admin test account if configured | Restricted data is inaccessible | Pending | Pending |

### 6.3 Debugging Deluge Scripts

The validation workflows were reported as updated with Deluge logic. Debugging should include reviewing the exact code in the workflow editor, checking field link names and lookup criteria, using the platform's available error messages or logs, and submitting both valid and invalid records. One reported builder issue during development was a Zoho Creator builder error involving missing lookup metadata; the Zia agent attributed it to a possible builder metadata issue. It should be mentioned as a development observation only if it was actually encountered in this app, and any final fix should be documented from the live environment.

### 6.4 Defect Log

**Table 6.2: Defect Log**

| Bug ID | Description | Root Cause | Fix Applied | Status |
|---|---|---|---|---|
| BUG-01 | Builder error reported in an earlier form-building stage: missing lookup reference metadata | Possible builder metadata/platform issue according to the builder assistant; root cause not confirmed | The forms were later reported as successfully created; exact repair steps not documented | Closed by continued build; verify no recurrence |
| BUG-02 | Validation workflows require actual negative-case testing | No test evidence supplied yet | Run duplicate-application and zero-salary tests | Open / pending testing |

### 6.5 User Acceptance Testing

The template requests feedback from at least two end users. No actual user feedback was supplied during preparation of this report. Complete the table after two users operate the application and provide genuine feedback.

**Table 6.3: User Acceptance Feedback**

| User | Role Tested | Feedback | Action Taken |
|---|---|---|---|
| [User 1 name] | [Role] | [Record actual feedback after test] | [Action taken] |
| [User 2 name] | [Role] | [Record actual feedback after test] | [Action taken] |

# CHAPTER 7

# DEPLOYMENT, SECURITY AND GOVERNANCE

## 7.1 Environments and Deployment Strategy

The application was configured in Zoho Creator. The conversation does not confirm that separate development, staging, and production environments were used or that the app was formally published to production. Before submission, confirm whether the app has a live URL and record the deployment date only if publication has occurred.

**Table 7.1: Deployment Checklist**

| S.No. | Checklist Item | Status |
|---|---|---|
| 1 | Forms, reports and workflows reviewed in the build environment | Reported created; manual verification required |
| 2 | Sample/test data removed or clearly isolated | Pending |
| 3 | Roles and permissions configured and tested | Not recorded |
| 4 | Connections authorised in production | Not applicable / no integrations confirmed |
| 5 | Workflows saved and enabled | Reported enabled; test behaviour |
| 6 | Application published and URL verified | Not recorded |
| 7 | Version / release notes recorded | Pending |

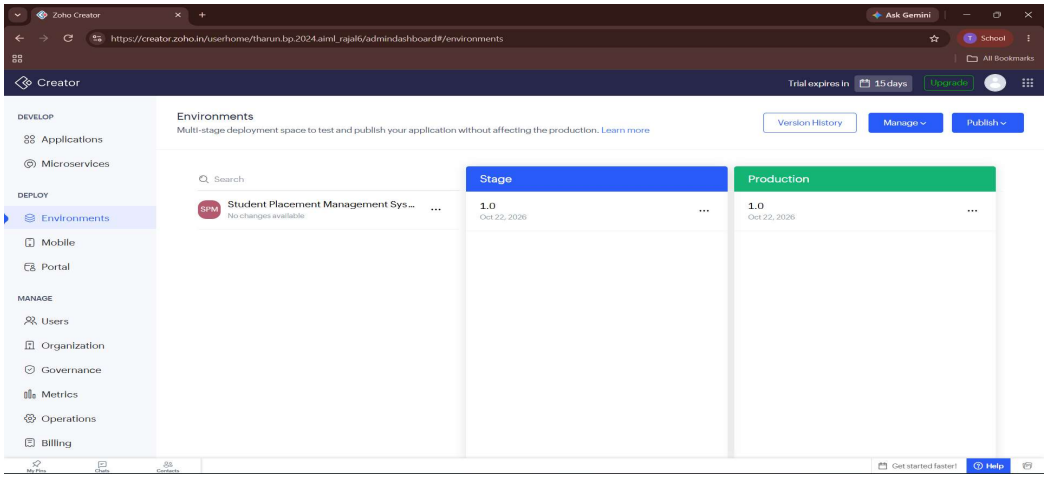



**Figure 7.1: Environments / Publish to Production**

### 7.2 Performance Optimisation

No measured performance benchmark was supplied. For a small student project, basic optimisation can include keeping reports focused on relevant fields, using appropriate filters, avoiding unnecessary repeated data lookups in custom Deluge code, and checking that dashboard cards use efficient data sources. Any claim of faster response time must be supported by before-and-after measurements.

**Table 7.2: Performance Optimisation Results**

| Optimisation Applied | Component | Before | After | Improvement |
|---|---|---|---|---|
| Use only relevant report columns | Reports | Not measured | Not measured | Not quantified |
| Use filtered views for relevant records | Reports | Not measured | Not measured | Not quantified |
| Review lookup/query logic in Deluge | Validation workflows | Not measured | Not measured | Not quantified |

### 7.3 Security Features

The application may contain personal and academic student data, so access should be limited to authorised users. Security must be verified through Creator permission settings and separate-account tests. Do not include demo passwords in the report. The current conversation did not confirm encryption settings, record-level criteria, two-factor authentication, or a student portal.

**Table 7.3: Security Controls Implemented**

| Security Control | Where Applied | Threat Mitigated / Status |
|---|---|---|
| Required fields and validation | Forms / validation workflows | Reduces incomplete or invalid records; test actual behaviour |
| Role-based permissions | Creator permissions | Not recorded; configure and test before real use |
| Record-level restrictions | Reports / data access | Not recorded; required for private student records |
| Secure integration connections | Integrations | No external integration confirmed |
| Credentials management | Application administration | Never include passwords in reports; use approved account controls |

### 7.4 Governance and Administration

Governance practices should include consistent form and workflow names, documenting changes, reviewing permissions, exporting or backing up data according to college requirements, and checking workflow logs after changes. The application owner should confirm which audit trail and backup capabilities are available under the current Creator edition.

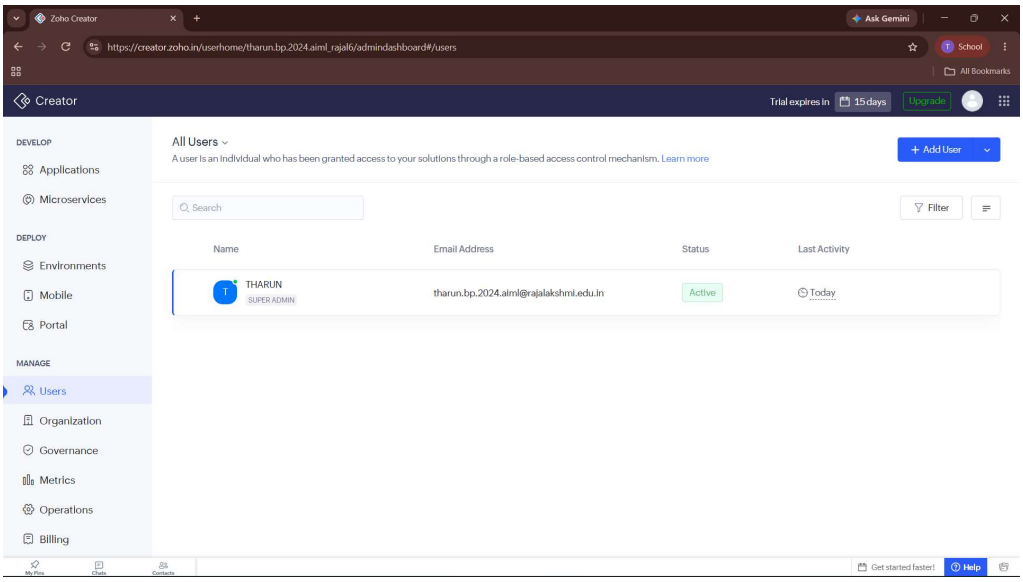



**Figure 7.2: Audit Trail / Administration Settings**

# CHAPTER 8
# RESULTS AND DISCUSSION

## 8.1 Application Output

The reported application output consists of nine forms, nine reports, a dashboard with six summary cards and three pie charts, and workflows for default statuses and validation. The end-to-end scenario below describes the intended demonstration path. Capture real screenshots and update the results after performing each step.

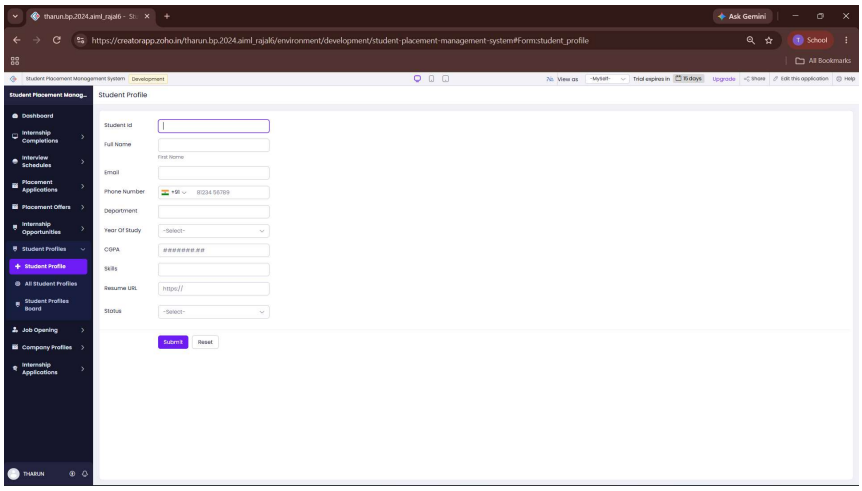



**Figure 8.1: Scenario Step 1 – Student, company, internship and job records**

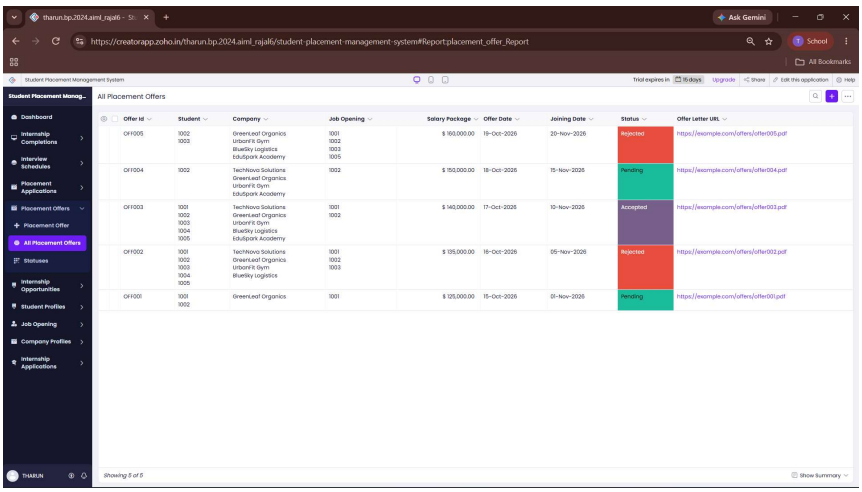



**Figure 8.2: Scenario Step 2 – Placement Offers**

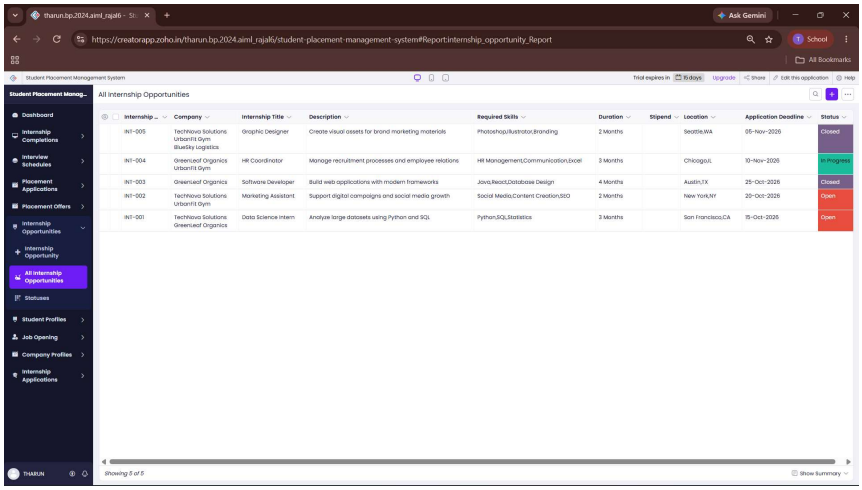



**Figure 8.3: Scenario Step 3 – Internship Opportunities**

## 8.2 Achievement of Objectives

**Table 8.1: Mapping of Objectives to Outcomes**

| Objective (Section 1.3) | Implemented Through | Outcome / Evidence |
|---|---|---|
| Create structured forms | Chapter 5.2 | Nine forms reported created; confirm |
| | | field definitions |
| Track applications | Chapter 5.2 and 5.3 | Internship and placement application forms/reports reported created |
| Maintain interviews, offers and completion | Chapter 5.2 and 5.3 | Forms and reports reported created |
| Provide reports | Chapter 5.3 | Nine reports reported created; live report content must be checked |
| Provide dashboard | Chapter 5.4 | Six cards and three pie charts reported created; verify against real records |

| | | |
|---|---|---|
| Automate defaults and validations | Chapters 5.5 and 5.6 | Workflows reported enabled; validation behaviour awaits negative-case tests |

### 8.3 Comparison with the Existing System

**Table 8.2: Existing System versus Proposed Application**

| Parameter | Existing / Manual Approach | Proposed Application | Improvement |
|---|---|---|---|
| Record storage | Separate files or informal records may be used | Central forms in one application | Centralisation intended; actual usage should be evaluated |
| Report generation | Manual filtering and consolidation | Reports are available in the app menu | Less manual navigation expected; not measured |
| Status tracking | Statuses updated in separate records or messages | Dropdown statuses and default workflows | More consistent status entry intended |
| Dashboard | Manual summaries may be required | Dashboard with cards and charts | Quick overview intended; accuracy needs testing |
| Data validation | Depends on manual checks | Validation workflows reportedly configured | Invalid submissions should be tested before claiming success |

### 8.4 Limitations of the Low-Code Approach

The current build depends on Zoho Creator's field types, report capabilities, workflow triggers, and available scripting features. Complex matching algorithms, extensive integrations, custom interfaces, and advanced access rules may require additional Deluge logic or a pro-code extension. AI-assisted configuration can accelerate setup but may create inaccurate field references or incomplete logic, so generated workflows must be inspected. A further limitation is that performance, security, and outcome improvements have not yet been measured.

### 8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Visual form creation, menu organisation, reports, and dashboard components are mainly no-code configuration. Workflow conditions and validation rules using Deluge are low-code. A separate recommendation engine, custom job-board connector, or specialised analytics service could require pro-code development. Moving to pro-code should be based on a real requirement that cannot be met efficiently within Creator, rather than adding complexity for its own sake.

# CHAPTER 9

# CONCLUSION AND FUTURE ENHANCEMENTS

## 9.1 Conclusion

The Student Internship and Placement Management System was configured in Zoho Creator to organise the main records involved in campus internship and placement activities. The reported build includes nine forms, nine reports, a dashboard with six summary cards and three pie charts, and workflows for setting default statuses and validating selected records. The application provides a structured starting point for centralising student, company, opportunity, application, interview, offer, and internship completion information.

The main remaining work is evidence-based verification. Real records should be entered, related records and report results checked, validation failures tested, and dashboard counts compared with the underlying data. Permissions, deployment, external integrations, and measured performance should only be reported as implemented after they have been configured and verified.

## 9.2 Future Enhancements

- Configure separate Administrator, Placement Officer, Recruiter, and Student access with record-level restrictions and test each role.
- Add email or in-app notifications for interview schedules, application decisions, and offer updates if supported by the selected Creator plan.
- Add scheduled reminders for application deadlines and upcoming interviews.
- Add a student portal for viewing open opportunities and personal application statuses.
- Integrate with a mail service or external job board using secure connections if required.
- Collect test metrics and user feedback, then use the results to improve form usability and report performance.
- Explore analytics or AI-assisted candidate matching only after data quality and governance are established.

### 9.3 Learning Outcomes and Career Relevance

This project provides practical experience with low-code application planning, form-based data capture, reports, dashboards, workflow configuration, and the need to verify AI-assisted output. It demonstrates how a citizen developer can deliver a useful prototype while recognising when custom scripting or pro-code development is necessary. These skills are relevant to low-code development, business process automation, solutions consulting, application administration, and software engineering roles.

# REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 19-Oct-2026.

[2] Zoho Corporation, “Deluge Script Reference.” [Online]. Available: https://www.zoho.com/deluge/. Accessed: 19-Oct-2026.

[3] Zoho Corporation, “Zoho Creator Resources.” [Online]. Available: https://www.zoho.com/creator/resources/. Accessed: 19-Oct-2026.

[4] Zoho Corporation, “Zoho Creator Workflow Documentation.” [Online]. Available through Zoho Creator Help. Accessed: 19-Oct-2026.

[5] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications. (Verify edition and publication year against the copy consulted before submission.)

# APPENDIX A

# ZOHO CREATOR FEATURE COVERAGE CHECKLIST

**Table A.1: Feature Coverage Checklist**

| Unit | Feature | Done? | Where in Application | Report Section / Evidence |
|---|---|---|---|---|
| I | Application created and menu organised | Reported; verify | Student Internship & Placement Management System | **5.1 / Figure 5.1** |
| II | Forms with at least 8 distinct field types | Not verified | Nine forms | **5.2 / Figures 5.2–5.3** |
| II | Lookup relationships between forms | Partially reported | Internship opportunity company lookup; other lookups need inspection | **4.2, 5.2** |
| II | Subform (parent–child data) | Not recorded | None confirmed | **4.2** |
| II | Field validations | Reported; test | Application and offer validation workflows | **5.5, 5.6** |
| II | List reports with filters/grouping/custom actions | Reports reported; details unverified | Nine reports | **5.3 / Figure 5.4** |
| II | Kanban/Calendar/ Timeline/Map report | Not recorded | Inspect report types | **5.3** |
| II | Pivot table/chart report | Pie charts on dashboard reported; report type not confirmed | Dashboard | **5.4 / Figure 5.5** |
| II | Page/dashboard built with UI Builder | Reported created | Student_Placement_Dashboa rd | **5.4 / Figure 5.5** |

| | | | | |
|---|---|---|---|---|
| II | Deluge form events / validation | Logic reported; source code to inspect | Three validation workflows | **5.5 / Appendix C** |
| II | Deluge functions, fetch, loops, collections | Not recorded | Inspect scripts | **5.5 / Appendix C** |
| II | Email/in-app notifications | Not recorded | None confirmed | **5.5–5.6** |
| II | Form workflows | Reported enabled | Default-status and validation workflows | **5.6 / Figure 5.6** |
| II | Scheduled workflow | Not recorded | None confirmed | **5.6** |
| II | Approval process | Not recorded | None confirmed | **5.6** |
| II | Blueprint | Not recorded | None confirmed | **5.6** |
| II | Batch/report workflow | Not recorded | None confirmed | **5.6** |
| II | Testing and debugging evidence | Pending | Test records/logs to be captured | Chapter 6 |
| III | Zoho service integration | Not recorded | None confirmed | **5.7** |
| III | Connection/REST API call | Not recorded | None confirmed | **5.7** |
| III | Webhook or Creator REST API | Not recorded | None confirmed | **5.7** |
| III | Users, roles and permission profiles | Not recorded | Inspect app permissions | **5.8** |
| III | Record-level access criteria / portal | Not recorded | Inspect app permissions/portal | **5.8, 5.11** |
| III | Zia / AI feature | Used for assisted configuration per build conversation | Zia Build Agent | **5.9** |
| III | Analytics (charts / Zoho Analytics) | Dashboard charts reported; Zoho Analytics integration | Student_Placement_Dashboard | **5.4, 5.9** |

| | | | | |
|---|---|---|---|---|
| | | not confirmed | | |
| IV | Performance optimisation applied and measured | Not measured | No benchmark supplied | **7.2** |
| IV | Deployment across environments | Not recorded | Inspect deployment settings | **7.1** |
| IV | Security controls (encryption, OAuth, permissions) | Not recorded | Inspect security settings | **7.3** |
| IV | Governance (audit trail, administration) | Not recorded | Inspect administration settings | **7.4** |
| V | Pro-code extension | Not implemented / not confirmed | None confirmed | **5.10** |
| V | Low-code limitations and spectrum analysis | Discussed | Chapter 8 | **8.4, 8.5** |
| — | Mobile app / portal access | Not recorded | None confirmed | **5.11** |

Total features implemented: do not calculate until each row is checked against the live application and supporting evidence.

# APPENDIX B

# COURSE OUTCOME MAPPING

**Table B.1: Mapping of Course Outcomes to Report Chapters**

| CO | Course Outcome Statement | Evidence in Report | Self-Assessed Level (1–3) | Examiner Level |
|---|---|---|---|---|
| CO1 | Explain principles, architecture and business value of low-code development using Zoho Creator. | Chapters 1 and 2 | [Select after review] | |
| CO2 | Develop business applications using Creator forms, reports, workflows and Deluge scripting. | Chapters 4, 5.1–5.6 and 6 | [Select after review] | |
| CO3 | Integrate Creator applications with Zoho services/external systems using APIs, connections and role-based security. | Sections 4.6, 5.7–5.9; integrations/RBAC not confirmed | [Select after review] | |
| CO4 | Deploy, secure and manage applications across environments and | Chapter 7; deployment and security require verification | [Select after review] | |

| | | | | |
|---|---|---|---|---|
| | governance practices. | | | |
| CO5 | Evaluate low-code strategies, AI-assisted development and transition to pro-code. | Sections 5.9, 5.10, 8.4, 8.5 and 9.3 | [Select after review] | |

# APPENDIX C

# COMPLETE DELUGE SOURCE CODE

The live Deluge source code was not provided in the conversation. Do not submit invented or illustrative code as if it were deployed. Open each workflow in Zoho Creator, copy its complete script, and paste it into the corresponding section below using Courier New 10 pt.

### C.1 Internship Application Validation

Purpose: Paste the complete script from the existing Internship Application Validation workflow.

```
// Paste verified source code here
// Include workflow name, form, trigger, and purpose.
```

### C.2 Placement Application Validation

Purpose: Paste the complete script from the existing Placement Application Validation workflow.

```
// Paste verified source code here
// Include workflow name, form, trigger, and purpose.
```

### C.3 Placement Offer Validation

Purpose: Paste the complete script from the existing Placement Offer Validation workflow.

```
// Paste verified source code here
// Include workflow name, form, trigger, and purpose.
```

### C.4 Default Status Workflows

Purpose: Paste scripts or document configuration for student Active, internship Applied, placement Applied, interview Scheduled, offer Pending, and completion In Progress workflows, as applicable.

```
// Paste verified source code here
// Include workflow name, form, trigger, and purpose.
```

# APPENDIX D

# USER MANUAL

## D.1 Accessing the Application

1. Open the Zoho Creator application URL provided by the application owner.
2. Sign in using your authorised Zoho account.
3. Open the Student Internship and Placement Management System from your applications.
4. Use the menu sections to open the required form, report, or dashboard.
5. Do not use another person's account or share passwords.

## D.2 Instructions for Administrator / Placement Officer

6. Open Student Management and add or review student records.
7. Open Company Management and add or review company records.
8. Open Opportunity Management to add internships and job openings.
9. Open Application Management to review internship and placement applications.
10. Open Interview Management to record interview details and feedback.
11. Open Placement Management to record offers and their statuses.
12. Open Internship Completion to record start/end dates, feedback, and certificate uploads.
13. Open the relevant report to verify saved records.
14. Open Student_Placement_Dashboard to review summary cards and pie charts; compare values against actual reports during testing.

## D.3 Instructions for Recruiter / Student

Recruiter and student-specific access has not been confirmed as configured. If separate roles or a portal are added, document the actual login path, permitted forms, record restrictions, and report access here. Until then, do not provide students or recruiters access to private records without verifying permissions.

# APPENDIX E

# EVALUATION RUBRIC (EXAMINER USE ONLY)

This appendix is retained from the prescribed template for examiner use. Students should not fill in marks awarded.

**Table E.1: Evaluation Rubric for the Individual Project**

| S.No. | Criterion | Indicators | Max. Marks | Marks Awarded |
|---|---|---|---|---|
| 1 | Problem Definition & Requirement Analysis | Problem clarity, objectives, FR/NFR, use cases | 10 | |
| 2 | System Design | Architecture, data model, form specifications, workflow design, permissions | 10 | |
| 3 | Forms, Reports & Pages | Field types, lookups/subforms, reports and dashboard | 15 | |
| 4 | Deluge Scripting & Workflows | Correctness of scripts and workflow depth | 15 | |
| 5 | Integration, RBAC & AI | Integrations, permissions, AI/analytics | 10 | |
| 6 | Testing, Deployment, Security & Governance | Test coverage, environments, optimisation, security | 10 | |
| 7 | Live Demonstration | End-to-end operation and | 15 | |

| | | | | |
|---|---|---|---|---|
| | | feature checklist | | |
| 8 | Project Report | Template adherence, format, figures, references | 5 | |
| 9 | Viva-Voce | Platform understanding and design decisions | 10 | |
| | TOTAL | | 100 | |

**Table E.2: Performance Level Descriptors**

| Level | % of Maximum Marks | Descriptor |
|---|---|---|
| Excellent | 90–100 | Feature implemented completely and correctly; design choices explained confidently. |
| Good | 70–89 | Feature implemented with minor gaps; explanation adequate. |
| Satisfactory | 50–69 | Feature partly implemented or works only for the main path. |
| Needs Improvement | Below 50 | Feature missing, non-functional or not understood. |

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name:_____

# APPENDIX F
# VIVA-VOCE RECORD SHEET (EXAMINER USE ONLY)

| | | | |
|---|---|---|---|
| Student Name | THARUN B P | Register No. | 241501230 |
| Department/Section | Artificial Intelligence and Machine Learning | Date of Viva | |
| Application URL Verified | Yes/No | Demo Duration | _____min |

**Table F.1: Viva-Voce Questions and Responses**

| Q.No. | Question / Topic Asked | Unit | Response (E/G/S/NI) | Remarks |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |

E – Excellent; G – Good; S – Satisfactory; NI – Needs Improvement.

**Table F.2: Marks Summary**

| Component | Maximum Marks | Marks Awarded |
|---|---|---|
| Criteria 1–8 (Appendix E) | 90 | |
| Viva-Voce | 10 | |
| TOTAL | 100 | |

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER
Name:_____

_____

EXTERNAL EXAMINER
Name:_____

## APPENDIX G: PUBLIC FORM LINK AND EMBED CODE

Company Registration Form – Permanent Link

https://creatorapp.zoho.in/tharun.bp.2024.aiml_rajal6/environment/stage/student-placement-management-system#Section_1

HTML Embed Code

<iframe height='500px' width='100%' frameborder='0' allowTransparency='true' scrolling='auto'src='https://creatorapp.zoho.in/tharun.bp.2024.aiml_rajal6/environment/stage/student-placement-management-system#Section_1'></iframe>

**Student:** Tharun B P | **Register Number:** 241501230 | **Platform:** Zoho Creator Student Edition